Examples of parameters that are structurally equivalent to θ_A (no assumption on initial or boundary conditions)

In[•]:=
```
(* Normalised so one coefficient is unity *)
coef =
 {l1 (l2 + l3 - l2 p12 p21 + l3 (-1 + p12) p31) + l2 l3 (p21 + p31 - p21 p31), l1 + l2 + l3,
  l1 l3 (1 + (-1 + p12) p31) v2 + l2 (l3 (p21 + p31 - p21 p31) v1 + l1 (1 - p12 p21) v3),
  l3 (v1 + v2) + l2 (v1 + v3) + l1 (v2 + v3), v1 + v2 + v3,
  l3 v1 v2 + l2 v1 v3 + l1 v2 v3, v2 v3 + v1 (v2 + v3), v1 v2 v3}
```

Out[•]=
```
{l1 (l2 + l3 - l2 p12 p21 + l3 (-1 + p12) p31) + l2 l3 (p21 + p31 - p21 p31), l1 + l2 + l3,
 l1 l3 (1 + (-1 + p12) p31) v2 + l2 (l3 (p21 + p31 - p21 p31) v1 + l1 (1 - p12 p21) v3),
 l3 (v1 + v2) + l2 (v1 + v3) + l1 (v2 + v3), v1 + v2 + v3,
 l3 v1 v2 + l2 v1 v3 + l1 v2 v3, v2 v3 + v1 (v2 + v3), v1 v2 v3}
```

In[•]:=
```
(*Extract only coefficients c7 and c8*)
othermeas = {l1 (l2 + l3 - l2 p12 p21 + l3 (-1 + p12) p31) + l2 l3 (p21 + p31 - p21 p31),
   l1 l3 (1 + (-1 + p12) p31) v2 + l2 (l3 (p21 + p31 - p21 p31) v1 + l1 (1 - p12 p21) v3)}
```

Out[•]=
```
{l1 (l2 + l3 - l2 p12 p21 + l3 (-1 + p12) p31) + l2 l3 (p21 + p31 - p21 p31),
 l1 l3 (1 + (-1 + p12) p31) v2 + l2 (l3 (p21 + p31 - p21 p31) v1 + l1 (1 - p12 p21) v3)}
```

In[•]:=
```
res = othermeas /. {v1 → 20, v2 → -15, v3 → 0, l1 → 1,
    l2 → 1 / 2, l3 → 3 / 10, p12 → 1 / 5, p21 → 3 / 10, p31 → 7 / 10}
```

Out[•]=
$$\left\{\frac{1441}{2000}, \frac{39}{100}\right\}$$

In[•]:=
```
neweqs = {othermeas[[1]] - res[[1]], othermeas[[2]] - res[[2]]}
```

Out[•]=
$$\left\{-\frac{1441}{2000} + l1\,(l2 + l3 - l2\,p12\,p21 + l3\,(-1 + p12)\,p31) + l2\,l3\,(p21 + p31 - p21\,p31),\ -\frac{39}{100} + l1\,l3\,(1 + (-1 + p12)\,p31)\,v2 + l2\,(l3\,(p21 + p31 - p21\,p31)\,v1 + l1\,(1 - p12\,p21)\,v3)\right\}$$

In[•]:=
```
examples = neweqs /. {v1 → 20, v2 → -15, v3 → 0, l1 → 1, l2 → 1 / 2, l3 → 3 / 10}
```

Out[•]=
$$\left\{\frac{159}{2000} - \frac{p12\,p21}{2} + \frac{3}{10}\,(-1 + p12)\,p31 + \frac{3}{20}\,(p21 + p31 - p21\,p31),\ -\frac{39}{100} - \frac{9}{2}\,(1 + (-1 + p12)\,p31) + 3\,(p21 + p31 - p21\,p31)\right\}$$

In[•]:=
```
sols = FullSimplify[Solve[0 == examples, {p21, p31}]]
```

Out[•]=

$$\left\{\left\{\text{p21} \to -\frac{1}{1000\,\text{p12}}\left(201+25\,\text{p12}\,(-71+30\,\text{p12})+\sqrt{40\,401+25\,\text{p12}\,(-24\,882+5\,\text{p12}\,(21\,701+300\,\text{p12}\,(-71+15\,\text{p12})))}\right),\right.\right.$$
$$\text{p31} \to -\frac{1}{1050\,(-1+\text{p12})}\left(849+25\,\text{p12}\,(-71+30\,\text{p12})+\sqrt{40\,401+25\,\text{p12}\,(-24\,882+5\,\text{p12}\,(21\,701+300\,\text{p12}\,(-71+15\,\text{p12})))}\right)\Big\},$$
$$\left\{\text{p21} \to \frac{1}{1000\,\text{p12}}\left(-201+25\,(71-30\,\text{p12})\,\text{p12}+\sqrt{40\,401+25\,\text{p12}\,(-24\,882+5\,\text{p12}\,(21\,701+300\,\text{p12}\,(-71+15\,\text{p12})))}\right),\right.$$
$$\left.\left.\text{p31} \to \frac{1}{1050\,(-1+\text{p12})}\left(-849+25\,(71-30\,\text{p12})\,\text{p12}+\sqrt{40\,401+25\,\text{p12}\,(-24\,882+5\,\text{p12}\,(21\,701+300\,\text{p12}\,(-71+15\,\text{p12})))}\right)\right\}\right\}$$

```
exA = FullSimplify[Solve[0 == examples /. {p12 → 2 / 10}, {p21, p31}]]
```

Out[•]=

$$\left\{\left\{\text{p21} \to \frac{3}{10},\ \text{p31} \to \frac{7}{10}\right\},\ \left\{\text{p21} \to \frac{47}{50},\ \text{p31} \to \frac{23}{42}\right\}\right\}$$

In[•]:=
```
exC = FullSimplify[Solve[0 == examples /. {p12 → 0}, {p21, p31}]]
```

Out[•]=

$$\left\{\left\{\text{p21} \to \frac{61}{268},\ \text{p31} \to \frac{108}{175}\right\}\right\}$$

In[•]:=
```
ex2 = FullSimplify[Solve[0 == examples /. {p12 → 1 / 100}, {p21, p31}]]
```

Out[•]=

$$\left\{\left\{\text{p21} \to \frac{1}{400}\left(-7333-\sqrt{55\,118\,569}\right),\ \text{p31} \to \frac{33\,253+\sqrt{55\,118\,569}}{41\,580}\right\},\right.$$
$$\left.\left\{\text{p21} \to \frac{1}{400}\left(-7333+\sqrt{55\,118\,569}\right),\ \text{p31} \to \frac{33\,253-\sqrt{55\,118\,569}}{41\,580}\right\}\right\}$$

In[•]:=
```
ex3 = FullSimplify[Solve[0 == examples /. {p12 → 5 / 100}, {p21, p31}]]
```

Out[•]=

$$\left\{\left\{\text{p21} \to \frac{1}{400}\left(-913-\sqrt{1\,008\,049}\right),\ \text{p31} \to \frac{6097+\sqrt{1\,008\,049}}{7980}\right\},\right.$$
$$\left.\left\{\text{p21} \to \frac{1}{400}\left(-913+\sqrt{1\,008\,049}\right),\ \text{p31} \to \frac{6097-\sqrt{1\,008\,049}}{7980}\right\}\right\}$$

In[•]:=
```
ex4 = Solve[0 == examples /. {p12 → 1 / 10}, {p21, p31}]
```

Out[•]=

$$\left\{\left\{\text{p21} \to \frac{1}{100}\left(-31-2\sqrt{679}\right),\ \text{p31} \to \frac{1}{945}\left(679+2\sqrt{679}\right)\right\},\right.$$
$$\left.\left\{\text{p21} \to \frac{1}{100}\left(-31+2\sqrt{679}\right),\ \text{p31} \to \frac{1}{945}\left(679-2\sqrt{679}\right)\right\}\right\}$$

In[ ]:=
```
ex4b = Solve[0 == examples /. {p12 → 0.1}, {p21, p31}]
```

Solve : Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

Out[ ]=
```
{{p21 → -0.831153, p31 → 0.773667}, {p21 → 0.211153, p31 → 0.66337}}
```

In[ ]:=
```
(* Here two solutions *)
ex5 = Solve[0 == examples /. {p12 → 2 / 10}, {p21, p31}]
```

Out[ ]=
$$\left\{\left\{p21 \to \frac{3}{10}, p31 \to \frac{7}{10}\right\}, \left\{p21 \to \frac{47}{50}, p31 \to \frac{23}{42}\right\}\right\}$$

In[ ]:=
```
ex6 = Solve[0 == examples /. {p12 → 3 / 10}, {p21, p31}]
```

Out[ ]=
$$\left\{\left\{p21 \to \frac{1}{100}\left(88+\sqrt{3399}\right), p31 \to \frac{1}{245}\left(128-\sqrt{3399}\right)\right\},\right.$$
$$\left.\left\{p21 \to \frac{1}{100}\left(88-\sqrt{3399}\right), p31 \to \frac{1}{245}\left(128+\sqrt{3399}\right)\right\}\right\}$$

In[ ]:=
```
ex6b = Solve[0 == examples /. {p12 → 0.3}, {p21, p31}]
```

Solve : Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

Out[ ]=
```
{{p21 → 1.46301, p31 → 0.284486}, {p21 → 0.296991, p31 → 0.760412}}
```

In[ ]:=
```
ex7 = Solve[0 == examples /. {p12 → 4 / 10}, {p21, p31}]
```

Out[ ]=
$$\left\{\left\{p21 \to \frac{1}{400}\left(389-\sqrt{69\,601}\right), p31 \to \frac{1}{630}\left(259+\sqrt{69\,601}\right)\right\},\right.$$
$$\left.\left\{p21 \to \frac{1}{400}\left(389+\sqrt{69\,601}\right), p31 \to \frac{1}{630}\left(259-\sqrt{69\,601}\right)\right\}\right\}$$

In[ ]:=
```
exb7 = Solve[0 == examples /. {p12 → 0.4}, {p21, p31}]
```

Solve : Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

Out[ ]=
```
{{p21 → 0.31295, p31 → 0.829873}, {p21 → 1.63205, p31 → -0.00765082}}
```

In[ ]:=
```
ex8 = Solve[0 == examples /. {p12 → 5 / 10}, {p21, p31}]
```

Out[ ]=
$$\left\{\left\{p21 \to \frac{1}{500}\left(499-2\sqrt{27\,469}\right), p31 \to \frac{1}{525}\left(149+2\sqrt{27\,469}\right)\right\},\right.$$
$$\left.\left\{p21 \to \frac{1}{500}\left(499+2\sqrt{27\,469}\right), p31 \to \frac{1}{525}\left(149-2\sqrt{27\,469}\right)\right\}\right\}$$

In[ ]:=
```
ex9 = Solve[0 == examples /. {p12 → 6 / 10}, {p21, p31}]
```

Out[ ]=
$$\left\{\left\{p21 \to \frac{1}{100}\left(99-\sqrt{3931}\right), p31 \to \frac{1}{70}\left(9+\sqrt{3931}\right)\right\},\right.$$
$$\left.\left\{p21 \to \frac{1}{100}\left(99+\sqrt{3931}\right), p31 \to \frac{1}{70}\left(9-\sqrt{3931}\right)\right\}\right\}$$

In[•]:=
```
ex10 = Solve[0 == examples /. {p12 → 7 / 10}, {p21, p31}]
```

Out[•]=

$$\left\{\left\{\text{p21} \to \frac{1}{700}\left(674 - \sqrt{155\,971}\right),\ \text{p31} \to \frac{1}{315}\left(-26 + \sqrt{155\,971}\right)\right\},\right.$$
$$\left.\left\{\text{p21} \to \frac{1}{700}\left(674 + \sqrt{155\,971}\right),\ \text{p31} \to \frac{1}{315}\left(-26 - \sqrt{155\,971}\right)\right\}\right\}$$

In[•]:=
```
ex11 = Solve[0 == examples /. {p12 → 8 / 10}, {p21, p31}]
```

Out[•]=

$$\left\{\left\{\text{p21} \to \frac{1}{800}\left(739 - \sqrt{146\,041}\right),\ \text{p31} \to \frac{1}{210}\left(-91 + \sqrt{146\,041}\right)\right\},\right.$$
$$\left.\left\{\text{p21} \to \frac{1}{800}\left(739 + \sqrt{146\,041}\right),\ \text{p31} \to \frac{1}{210}\left(-91 - \sqrt{146\,041}\right)\right\}\right\}$$

In[•]:=
```
ex11 = Solve[0 == examples /. {p12 → 9 / 10}, {p21, p31}]
```

Out[•]=

$$\left\{\left\{\text{p21} \to \frac{1}{300}\left(263 - 2\sqrt{2941}\right),\ \text{p31} \to \frac{1}{35}\left(-47 + 2\sqrt{2941}\right)\right\},\right.$$
$$\left.\left\{\text{p21} \to \frac{1}{300}\left(263 + 2\sqrt{2941}\right),\ \text{p31} \to \frac{1}{35}\left(-47 - 2\sqrt{2941}\right)\right\}\right\}$$